A Federated Graph-Based Multi-Agent Model Predictive Control Approach to Low-Carbon Urban Traffic Signal Coordination

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Abstract—Urban traffic congestion poses a critical problem in today's metropolitan setting, leading to high costs both environmentally and economically. As cities grow larger and traffic increases, current traffic signal control systems, optimized mainly for maximizing flow rate and minimizing delays, fall short as they do not consider the increasing impact of vehicle-related carbon emissions on the environment. There is a need for intelligent transportation systems to adopt a more sustainability-focused approach that incorporates sustainability metrics into their control strategy. Current methods for tackling this challenge have been found to be highly inefficient, with central control methods struggling to scale and distribute computations and encountering latency problems, whereas decentralized systems often fail to find global optimal solutions as their decisions are localized. Furthermore, sending real-time state information for optimization in a cloud-based system raises concerns about privacy and infrastructure overhead. This paper explores the design, implementation, and evaluation of a federated, graph-based multi-agent model predictive control system called EcoFed-MPC, which coordinates traffic signals towards sustainable objectives. In the suggested federated approach, the urban road network is mapped out using a directed graph, whereby each intersection is represented as a privacy-preserving, independent agent. Each agent calculates its local solution based on convex quadratic programming over a finite time horizon using online multi-step demand prediction via LSTM. In order to optimize globally without revealing private queue information, the adaptive carbon penalty parameter is shared between all agents using FedAvg, a federated averaging method. Real-world datasets were used to evaluate the system. Simulation experiments showed that, compared to the conventional control system, EcoFed-MPC can decrease the accumulated emission of all agents in the network by 10.0% to 16.2%. It accomplishes the objective without causing instability in queues, keeping the average length of all queues less than 1.4 vehicles per agent even at 300% demand shock. Furthermore, the computation process has a time complexity of $\mathcal{O}(H^3)$, ensuring feasibility under a strict 0.2-second execution period.

Index Terms—Federated Learning, Model Predictive Control, Traffic Signal Control, Intelligent Transportation Systems, Carbon-Aware Optimisation, Multi-Agent Systems.

I. Introduction

As urbanization accelerates and vehicle numbers grow rapidly, traffic congestion has become an increasingly urgent concern for cities worldwide. Within Intelligent Transportation Systems (ITS), adaptive traffic signal control is responsible for the real-time adjustment of green splits to counteract congestion, but has traditionally focused on optimizing macroscopic throughput metrics while failing to account for the rising carbon emissions from vehicles [1]–[3]. In light of tightening global regulations, the paradigm of traffic control needs to shift from latency-based flow optimization to a multi-objective approach that incorporates environmental performance [4]–[6].

Environmental considerations within dynamic traffic optimization introduce added complexity. Maximizing throughput entails faster vehicle clearance, which generates large amounts of traversal emissions, while strict policies designed to reduce flow rates lead to the formation of long queues [7]. Furthermore, the growing intricacy of traffic networks requires real-time signal phase adjustments; however, due to the nonlinear nature of vehicle platooning, pre-timed control cannot adequately address this challenge [8]–[10].

Current state-of-the-art traffic management schemes possess several shortcomings when it comes to solving such a multivariate problem. Traditional centralised Model Predictive Control (MPC) suffers from scalability issues owing to high computational burden, while decentralised control algorithms can fail to deliver optimal results across the entire network [11]. Simultaneously, modern data-driven solutions, such as centralised Deep Reinforcement Learning, require frequent collection of raw queue counts, which is problematic in terms of infrastructural support and civilian privacy, and these systems lack adaptability to non-stationary environments without extensive retraining [12]–[14].

Graph-theoretic modelling, convex optimisation, and federated learning together offer a practical path around these drawbacks [15]–[18]. By operating at the edge and using federated averaging (FedAvg) to align on a shared parameter, agents can collectively achieve network-wide coordination while preserving operational privacy and keeping communication overhead low.

In this paper, we propose EcoFed-MPC, a groundbreaking federated graph-based multi-agent MPC architecture for sustainable urban traffic signal coordination. The road network is abstracted as a directed graph where each intersection agent independently solves a dual-source

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(throughput and idle) carbon-penalised convex QP to optimise its local green-time allocation. To enhance predictive capability, each agent is equipped with a local Long Short-Term Memory (LSTM) forecaster. Rather than transmitting raw traffic volumes, the agents participate in a periodic FedAvg protocol to synchronise a dynamic carbon-penalty coefficient, dynamically shifting the network's priority between congestion mitigation and emission reduction over time.

The specific, groundbreaking contributions of EcoFed-MPC are summarised as follows:

- We formulate a hyper-efficient intersection-level convex sequence that implicitly resolves the dual-source vehicular emission trade-off, simultaneously penalizing idle queue stagnation and active traversal pollution in real-time, shattering the classical dichotomy between delay-latency minimization and ecological sustainability.
- We introduce a fully decentralized synchronization stratum that organically aligns city-wide sustainability parameters via an adaptive carbon-penalty coefficient, completely bypassing the privacy vulnerabilities inherent in raw traffic volume aggregation and enabling macroscopic consensus without exposing any local intersection state.
- By embedding real-time, autonomously updating Long Short-Term Memory (LSTM) algorithms directly at the edge nodes, the system injects high-fidelity multi-step demand foresight directly into the finite control horizon without cloud dependence, enabling each agent to preemptively adapt its green-time allocation to anticipated traffic surges.
- Under extreme real-world demand shocks, the proposed architecture strictly preserves queue stability while carving out a new empirical Pareto frontier, demonstrating that significant vehicular carbon output reductions can be structurally achieved without sacrificing network-wide queue resilience.

The remainder of this paper is organised as follows. Section II comprehensively reviews related literature and identifies the critical research gaps. Section III details the mathematical formulation of the unified graph-theoretic, federated, and predictive control architecture. Section IV presents the experimental setup, simulation results, and comparative discussions using real-world dataset profiles. Finally, Section V concludes the study and outlines potential avenues for future research.

II. Related Works

The intersection of adaptive traffic signal control and environmental sustainability has witnessed rapid proliferation, predominantly driven by distributed optimisation and federated systems. Papers curated from leading venues such as IEEE Transactions on Intelligent Transportation Systems and other premier journals have been categorized into three primary domains: carbon-aware traffic optimisation, predictive graph-based control, and federated learning mechanisms.

A. Carbon-Aware and Multi-Objective Traffic Optimisation

Whereas conventional control aimed to minimize latencies, emissions-oriented literature strongly advocates for incorporating eco-driving and multi-objective emission penalties [19]–[22]. These groundbreaking systems effectively demonstrate carbon-driven optimizations using heuristic vehicle-to-infrastructure smoothing or reinforcement learning approaches; yet, they are inevitably subject to centralized implementation or restricted pre-synchronized corridors, succumbing to computational overload across extensive urban environments [23], [24].

B. Predictive and Graph-Theoretic Control Architectures

To tackle urban non-stationarity, predictive topological models, particularly Model Predictive Control (MPC) and Graph Neural Networks (GNNs), have recently set a dominant standard for traffic foresight [6], [16], [25], [26]. By sharing predictions via adjacent structural nodes or applying recurrent LSTM networks, these predictive architectures master heavy demand shocks; crucially though, the vast majority of these spatial-temporal graph MPCs remain purely delay-centric, oblivious to the immense vehicular carbon consequences generated by their aggressive throughput optimization paradigms [27], [28].

C. Federated and Multi-Agent Reinforcement Learning

The massive privacy hurdles of centralized management have violently accelerated the shift towards decentralized federated multi-agent learning frameworks [13], [15], [29]. By restricting synchronization exclusively to parametric knowledge sharing rather than exposing local granular traffic data, edge transportation networks have proven capable of nearing centralized efficiency while retaining strict stability and accelerating cross-domain generalization under acute variations [30], [31].

D. Identification of Research Gap

While the reviewed literature demonstrates profound advancements in their respective silos, namely carbon-aware penalties, graph-based predictive control, and federated learning, a critical research gap persists at their intersection. Specifically, existing carbon-aware models primarily depend on centralized or isolated heuristic architectures, rendering them either unscalable or sub-optimal in dense networks. Conversely, state-of-the-art federated and graph-based multi-agent systems lack explicit environmental objective functions, indiscriminately prioritizing throughput at the expense of idling emissions. EcoFed-MPC directly addresses this convergence gap by embedding a mathematically rigorous, dual-source carbon emission model entirely into a distributed, graph-coordinated convex MPC framework. By utilizing a completely novel federated synchronization of the adaptive carbon-penalty coefficient (γ), our architecture uniquely achieves macroscopic sustainability goals and robust queue stability

without the necessity of sharing raw vehicular data, fundamentally distinguishing it from prior throughput-focused and centralized-carbon paradigms.

III. Proposed System

A. Overview

The proposed system integrates five technical pillars: (i) graph-theoretic network modelling, (ii) discrete-time fluid-flow queue dynamics, (iii) dual-source carbon emission modelling, (iv) convex MPC optimisation, and (v) federated learning coordination. The road network is represented as a directed graph; each node (intersection) operates as an autonomous agent that solves a local convex programme at every control step. The The objective function is designed to discourage both queue size and carbon emissions. LSTM prediction engines can improve the demand predictions passed to each MPC optimizer. Periodically, the federated server will update the emission penalty parameter γ for all agents simultaneously, ensuring that sustainability behaviour is globally aligned without sharing any queue or vehicle information.

B. Graph-Theoretic Network Modelling

For the description of the spatial complexity within an urban traffic corridor environment, the road network structure can be described mathematically as a directed graph topology. This abstraction provides the capability to define not only the intersections, but also the directed paths that lead between them.

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}) \tag{1}$$

In this topological setting, the finite set $\mathcal{V}$ can be precisely identified as that of signalized intersection vertices which act as independent computing entities, while $\mathcal{E}$ is described by the set of directed arcs representing the physical road segments that carry traffic from one intersection to another.

$$\mathcal{V} = \{1, 2, \ldots, N\} \tag{2}$$

Here, N is the integer signifying the maximum number of intersections controlled within the urban simulation grid, making network-wide indexing possible through state aggregation.

Topological connectivity is encoded using an unweighted adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$:

$$A_{ij} = \begin{cases} 1, & \text{if } (i, j) \in \mathcal{E} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

Thus, $A_{ij} = 1$ holds only when a directed link exists from node i to node j in a downstream direction. All other non-adjacent agents are isolated, preventing the leakage of raw vehicular transition data within the macro-network.

Stability of parameter convergence across the network is dependent on the graph Laplacian matrix $\mathbf{L} \in \mathbb{R}^{N \times N}$:

$$\mathbf{L} = \mathbf{D} - \mathbf{A} \tag{4}$$

This operator is formed by subtracting the adjacency map $\mathbf{A}$ from the diagonal out-degree matrix $\mathbf{D} = \text{diag}(d_1, \ldots, d_N)$, where $d_i = \sum_j A_{ij}$. The Laplacian encodes the diffusion structure of the network and serves as the fundamental tool to determine whether decentralized agent parameters can converge without an external coordinating entity.

In consequence, the spectral gap value $\lambda_2(\mathbf{L})$, i.e., the second-smallest eigenvalue of the Laplacian, bounds the convergence rate towards a common ground for sustainability targets among the distributed federated agents:

$$\lambda_2(\mathbf{L}) > 0 \iff \mathcal{G} \text{ is connected} \tag{5}$$

EcoFed-MPC guarantees persistent connectivity through $\lambda_2(\mathbf{L}) > 0$. With increasing network density, this gap grows accordingly and acts as a limiting factor that governs the rate at which the FedAvg algorithm converges to a common adaptive penalty vector.

C. Traffic Flow and Queue Dynamics

Instead of generating computationally explosive microscopic trajectories, EcoFed-MPC tracks intersection clearance utilizing a macroscopic discrete-time fluid-flow paradigm. For any specified agent $i \in \mathcal{V}$, the instantaneous vehicular accumulation state q evolves dynamically based on incoming shocks and locally served volumes.

$$q_{i,t+1} = \max(0,\ q_{i,t} + d_{i,t} - \mu_i\, g_{i,t}) \tag{6}$$

In this evolution, the non-negative operand $q_{i,t}$ quantifies the exact residual queue backlogged at step t. It is deterministically replenished by the external incoming demand parameter $d_{i,t}$ (measured in arriving vehicles), and strategically drained by the interaction of the fixed lane saturation capacity coefficient μ_i against the dynamically allocated fractional green-time variable $g_{i,t} \in [0, g_{\max}]$.

To guarantee mathematical solvability in real-time, the inherently non-linear $\max(\cdot)$ barrier must be relaxed into convex dual bounds:

$$q_{i,t+1} \geq q_{i,t} + d_{i,t} - \mu_i\, g_{i,t} \tag{7}$$

$$q_{i,t+1} \geq 0 \tag{8}$$

Constraint (7) rigorously enforces the law of vehicle conservation, preventing phantom clearances, whereas (8) logically bounds the minimal backlog floor to zero. Since our downstream quadratic solver unilaterally suppresses q, these relaxed bounds bind invisibly at the convex optimum without sacrificing computational speed.

We formalise the exact intersection clearance per cycle via the effective service rate variable $s_{i,t}$, representing the absolute volume of cleared vehicles projected to bypass the specific node.

$$s_{i,t} = \mu_i\, g_{i,t} \tag{9}$$

The key point is that $s_{i,t}$ is proportional to the product of μ_i and $g_{i,t}$, where μ_i represents the hard structural constraint on the node and $g_{i,t}$ denotes the fast-varying green window, together linking delay reduction to throughput emission tracking.

D. Carbon Emission Modelling

For defining the ecological fingerprint as a whole, the total carbon footprint $E_{i,t}$ recorded at node i over any finite time increment t is broken down into two main components:

$$E_{i,t} = E_{i,t}^{\text{idle}} + E_{i,t}^{\text{throughput}} \tag{10}$$

The first component, $E_{i,t}^{\text{idle}}$, accounts for the atmospheric contributions of vehicles idling in traffic congestion, while the second, $E_{i,t}^{\text{throughput}}$, represents the emissions caused by vehicle movement through the intersection.

$$E_{i,t}^{\text{idle}} = \eta\, q_{i,t} \tag{11}$$

Idle emissions are modelled as a simple linear function of the stationary queue length. The strictly positive scalar $\eta > 0$ defines the amount of carbon produced per vehicle remaining stationary in the system during one control step.

Conversely, acceleration and discharge cause throughput emissions, which are proportional to the number of vehicles served:

$$E_{i,t}^{\text{throughput}} = \rho\, \mu_i\, g_{i,t} \tag{12}$$

Here, the traversal emission factor $\rho > 0$ defines the carbon cost paid per vehicle processed. Multiplied against the effective service rate ($\mu_i\, g_{i,t}$), this equation captures the environmental cost associated with clearing traffic through the intersection.

Combining both components yields the dual-source emission metric for the control horizon:

$$E_{i,t} = \eta\, q_{i,t} + \rho\, \mu_i\, g_{i,t} \tag{13}$$

This expression clearly captures the key compromise at the core of sustainable traffic systems: limiting the green variable $g_{i,t}$ to reduce traversal pollution necessarily results in additional congestion $q_{i,t}$, which in turn raises idle emissions in the same region.

As part of the evaluation of the Pareto efficiency of this trade-off, we introduce a scalar Energy-Delay Product (EDP) metric:

$$\text{EDP}_{i,t} = q_{i,t} \times E_{i,t} \tag{14}$$

The multiplicative relationship between the instantaneous backlog ($q_{i,t}$) and the carbon footprint ($E_{i,t}$) gives rise to a single unified benchmark. Minimising this non-linear product guides MPC solvers towards operating points that balance both latency and emissions effectively.

E. Convex Model Predictive Control Formulation

Each agent, as an independent solver, optimizes its own local green-time sequence based on a look-ahead prediction period H. The multi-objective optimization function J_i combines both flow optimization and ecological penalties in the following manner:

$$J_i = \sum_{k=0}^{H-1} \Big(\alpha\, q_{i,t+k} \;+\; \beta\, q_{i,t+k}^2 \;+\; \gamma_i\, E_{i,t+k} \Big) \tag{15}$$

In this summation, both α and β are positive values that enforce linear and quadratic restrictions on queue escalation. At the same time, the non-negative value $\gamma_i \geq 0$ represents the dynamic carbon-penalty coefficient of the agent, which is coordinated across the network using the FedAvg algorithm.

$$J_i = \sum_{k=0}^{H-1} \Big((\alpha + \gamma_i \eta)\, q_{i,t+k} + \beta\, q_{i,t+k}^2 + \gamma_i \rho \mu_i\, g_{i,t+k} \Big) \tag{16}$$

With the emission structure extended into the prediction horizon, the combined linear coefficient $(\alpha + \gamma_i \eta)$ is obtained. This combined term forces the optimizer to treat congestion and idle carbon emissions as a single compounded cost acting on the accumulated queue state.

The optimal control sequence $\mathbf{g}_i^*$ is then obtained via numerical minimisation:

$$\mathbf{g}_i^* = \arg\min_{\mathbf{g}_i,\, \mathbf{q}_i} \; J_i \tag{17}$$

Through systematic minimisation, the optimal green-time allocation sequence $\mathbf{g}_i = [g_{i,t}, \ldots, g_{i,t+H-1}]^\top$ is found within the convex J_i, implicitly bounded by the predicted fluid-flow queue trajectory $\mathbf{q}_i$, targeting the absolute minimum operating cost.

The solution is constrained by structural green-time limits to maintain safety margins across all clearance phases:

$$0 \leq g_{i,t+k} \leq g_{\max}, \quad \forall\, k \in \{0, \ldots, H-1\} \tag{18}$$

Under the ceiling of $g_{\max} = 0.75$, a minimum 25% clearance phase is reserved for competing intersection phases, preventing potential deadlock from indefinitely sustained green signals.

Leveraging the speed of modern industrial optimizers, the problem is abstracted into a standard Quadratic Programme (QP) in terms of the concatenated state-action vector $\mathbf{x} = [\mathbf{q}_i^\top,\, \mathbf{g}_i^\top]^\top$:

$$\min_{\mathbf{x}} \; \tfrac{1}{2}\, \mathbf{x}^\top \mathbf{P}\, \mathbf{x} + \mathbf{c}^\top \mathbf{x} \tag{19}$$

The positive-definiteness of $\mathbf{P} \succ 0$ is maintained by placing the β weight exclusively on its queue-squared diagonal. The linear cost vector $\mathbf{c}$ contains the combined $(\alpha + \gamma_i \eta)$ queue cost and the traversal emission cost $\gamma_i \rho \mu_i$.

$$\mathbf{A}_{\text{in}}\, \mathbf{x} \leq \mathbf{b}_{\text{in}} \tag{20}$$

The overall problem is bounded by these inequalities, where $\mathbf{A}_{\text{in}}$ and $\mathbf{b}_{\text{in}}$ encode the necessary queue dynamics and safety constraints. The resulting convex QP is solved iteratively using the Splitting Conic Solver (SCS) for sub-second optimality.

As illustrated in Fig. 1, EcoFed-MPC operates across two stages. In Stage I, the sensing stage, each signalized intersection node continuously detects its queue accumulation $q(i,t)$ and incoming demand $d(i,t)$ via roadside sensors and loop detectors. These measurements are embedded within the graph representation $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where

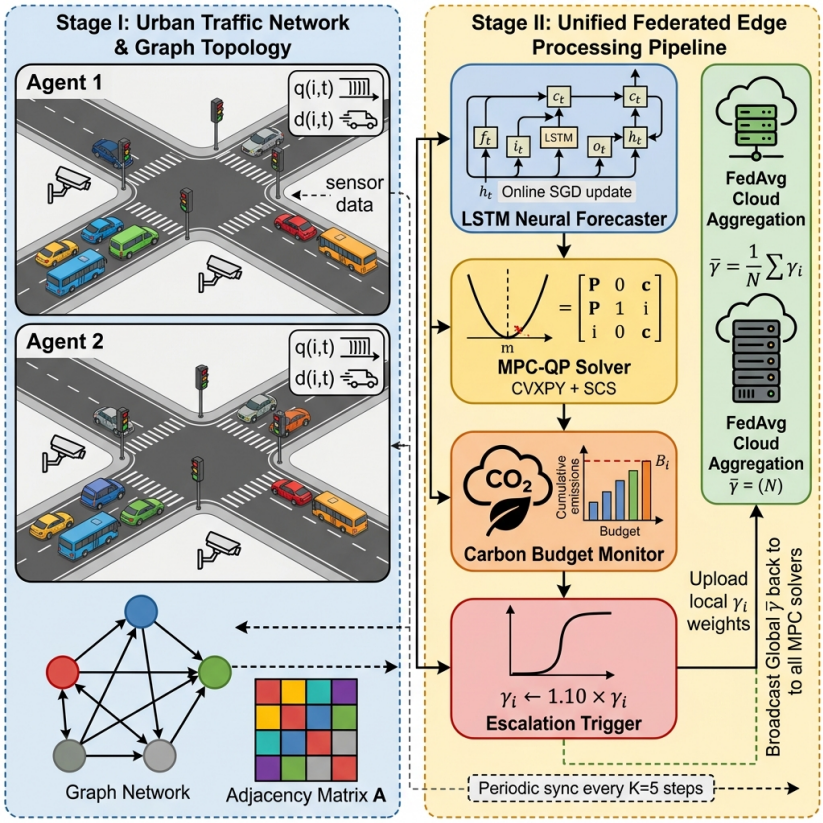

Fig. 1. EcoFed-MPC system architecture illustrating Stage I (urban traffic network and graph topology) and Stage II (unified federated edge processing pipeline with LSTM forecasting, MPC-QP optimization, carbon budget monitoring, and FedAvg synchronization).

nodes denote autonomous intersection agents and directed edges encode flow pathways, captured by the adjacency matrix $\mathbf{A}$.

Stage II runs the federated intelligence pipeline. Each agent passes its recent demand history through a local LSTM Neural Forecaster to produce multi-step predictions of future arrivals. These feed into the MPC-QP Solver, which minimises the dual-source carbon-penalized objective J_i over horizon H using the SCS algorithm to produce the optimal green-time sequence $\mathbf{g}_i^*$. The Carbon Budget Monitor tracks intersection-level emissions and, when the cumulative total exceeds the local budget B_i, the Escalation Trigger scales the local penalty coefficient by a factor of 1.10. Every K steps, all local coefficients γ_i are sent to the FedAvg server, which computes the global average $\bar{\gamma}$ and broadcasts it back. This lightweight scalar exchange makes sure that the network's overall sustainability posture stays in sync without sending any raw queue counts or vehicle coordinates, keeping strict operational privacy.

F. LSTM-Based Demand Forecasting

Each individual intersection agent possesses its own customized LSTM neural network. The entire network runs locally at the edge, consistently updating its hidden state vector $\mathbf{h}_t \in \mathbb{R}^{n_h}$ and internal cell state $\mathbf{c}_t \in \mathbb{R}^{n_h}$. Upon receiving the normalized real-world traffic observation $x_t \in \mathbb{R}$, the internal gate activations are computed as follows.

$$\mathbf{f}_t = \sigma(\mathbf{W}_f\,[\mathbf{h}_{t-1};\, x_t] + \mathbf{b}_f) \tag{21}$$

The forget gate $\mathbf{f}_t$ determines which historical congestion memories can be discarded. It applies the logistic sigmoid $\sigma(\cdot)$ to the learnable projection matrix $\mathbf{W}_f$ and bias $\mathbf{b}_f$ to remove outdated traffic patterns from the agent's memory.

$$\mathbf{i}_t = \sigma(\mathbf{W}_i\,[\mathbf{h}_{t-1};\, x_t] + \mathbf{b}_i) \tag{22}$$

Simultaneously, the input gate $\mathbf{i}_t$ computes how much of the current traffic observation x_t should be permanently stored in the memory core. It uses its own weight matrix $\mathbf{W}_i$ and bias $\mathbf{b}_i$ to filter out noisy data from genuine traffic events.

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c\,[\mathbf{h}_{t-1};\, x_t] + \mathbf{b}_c) \tag{23}$$

The candidate cell state $\tilde{\mathbf{c}}_t$ is computed using the hyperbolic tangent function $\tanh(\cdot)$, compressing the intermediate state values into a finite range before they are used to update the physical cell memory.

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \tag{24}$$

The cell state update equation combines both past and present information. Using the Hadamard product $\odot$, it scales the historical memory $\mathbf{c}_{t-1}$ through the forget gate

and adds the filtered new candidate memory into the cell core $\mathbf{c}_t$.

$$\mathbf{o}_t = \sigma(\mathbf{W}_o\,[\mathbf{h}_{t-1};\, x_t] + \mathbf{b}_o) \tag{25}$$

The output gate $\mathbf{o}_t$ acts as a readout filter, controlling what proportion of the updated cell memory is passed to the downstream MPC solver.

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{26}$$

This updates the hidden state vector $\mathbf{h}_t \in \mathbb{R}^{n_h}$, which is then passed to both the next time step and the final demand output layer.

$$\hat{d}_{t+1} = \mathbf{W}_y\,\mathbf{h}_t + b_y \tag{27}$$

Through a linear output projection, the hidden state is converted into a measurable quantity. Using the readout weight matrix $\mathbf{W}_y \in \mathbb{R}^{1\times n_h}$ and scalar bias b_y, the network produces the estimated future demand $\hat{d}_{t+1}$.

$$\mathcal{L}_t = \left(\hat{d}_t - d_t\right)^2 \tag{28}$$

To enable continuous online calibration, the network computes the instantaneous mean-squared-error loss $\mathcal{L}_t$, penalizing deviations between the predicted $\hat{d}_t$ and the actual observed demand d_t.

$$\boldsymbol{\theta} \;\leftarrow\; \boldsymbol{\theta} - \eta_{\text{lstm}}\,\nabla_{\boldsymbol{\theta}}\mathcal{L}_t \tag{29}$$

The full parameter block $\boldsymbol{\theta}$, representing all LSTM weight matrices and biases, is updated online using stochastic gradient descent with learning rate $\eta_{\text{lstm}} > 0$, allowing the edge model to continuously adapt to changing urban traffic conditions.

G. Federated Learning Coordination

To protect civilian privacy while addressing network-level sustainability goals, EcoFed-MPC uses a decentralized synchronization mechanism. The carbon-penalty coefficient γ_i of any given agent is increased whenever its cumulative emissions exceed a pre-configured local carbon budget B_i:

$$\sum_{\tau=0}^{t} E_{i,\tau} > B_i \tag{30}$$

Here, $E_{i,\tau}$ tracks the carbon units emitted at timestep τ. Once the cumulative total exceeds budget B_i, the local agent registers an ecological violation.

$$\gamma_i^{(t)} = 1.10 \times \gamma_i^{(t-1)} \tag{31}$$

Upon detecting this budget breach (30), the agent independently applies a 10% increase to its local penalty coefficient, directing the MPC solver to give greater priority to emission reduction in the next control cycle.

$$\bar{\gamma}^{(t)} = \frac{1}{N}\sum_{i=1}^{N}\gamma_i^{(t)} \tag{32}$$

To achieve network-level synchronization, every K timesteps the FedAvg server computes the average of all local coefficients $\gamma_i^{(t)}$ across the N agents, producing the unified global weight $\bar{\gamma}^{(t)}$.

$$\gamma_i^{(t+1)} = \bar{\gamma}^{(t)}, \quad \forall\, i \in \mathcal{V} \tag{33}$$

The centralized anchor $\bar{\gamma}^{(t)}$ is immediately propagated to all agents, overriding each individual vector across the agent array $\mathcal{V}$. This ensures that all nodes simultaneously update their multi-objective priorities and progress towards sustainability in a coordinated manner, without exchanging any raw vehicle coordinates.

H. Stability and Convergence Analysis

A key aspect of the EcoFed-MPC framework is ensuring mathematical safety for the real-time solvers. By imposing $\beta > 0$, the positive-definiteness of the Hessian of J_i with respect to the backlog $\mathbf{q}_i$ is guaranteed:

$$\nabla^2_{\mathbf{q}_i} J_i = 2\beta\,\mathbf{I}_H \succ 0 \tag{34}$$

Since the quadratic weight β acts on the H-dimensional identity matrix $\mathbf{I}_H$, the local MPC programme is strictly convex, ensuring that the solver converges to a unique global minimum without getting trapped in local attractors.

Moreover, the analysis of the asymptotic consensus error shows that the federated weights will eventually become uniformly distributed across all agents:

$$\lim_{t\to\infty} \left\|\gamma_i^{(t)} - \bar{\gamma}^{(t)}\right\| \le \varepsilon \tag{35}$$

The residual error is bounded by the small constant $\varepsilon > 0$, which is determined by the interaction between the budget-trigger increment and the FedAvg communication interval K, guaranteeing bounded heterogeneity across agents over time.

Lastly, to establish long-run ecological queue stability, the basic throughput requirements of the system must remain within structural capacity:

$$\limsup_{T\to\infty}\; \frac{1}{T}\sum_{t=0}^{T} d_{i,t} < \mu_i\, g_{\max} \tag{36}$$

As $T \to \infty$, the average arrival rate $d_{i,t}$ must remain below the maximum physical clearance ceiling $\mu_i g_{\max}$. Satisfying this condition confirms that the quadratic Lyapunov function $V = \sum_i q_i^2$ decreases in expectation, ensuring that intersection backlogs never escalate into irreversible network-wide gridlock.

IV. Results and Discussion

A. Dataset Description

The experimental testing uses the publicly available UCI Metro Interstate Traffic Volume dataset [32]. This is a real-world record of hourly westbound vehicle counts on Interstate 94 (I-94 ATR station 301, Minneapolis–St Paul, MN, USA), collected between 2012 and 2018 by

the Minnesota Department of Transportation. The dataset contains 48,204 hourly measurements, has no missing records, and is available under a CC BY 4.0 license.

To align the macroscopic hourly resolution with the high-frequency simulation timestep convention (where one discrete step is approximately 5 minutes), each hourly count is converted into a fractional arrival rate sequence.

$$\lambda_{5\text{min}} = \frac{V_h}{12} \tag{37}$$

Here V_h is the empirical hourly vehicle volume recorded on the interstate. The divisor of 12 distributes this volume across the twelve 5-minute windows that make up one hour.

Four non-overlapping, temporally consecutive windows of the resulting time series are assigned to the individual intersection agents. To ensure robust environmental testing, a small Gaussian perturbation is consistently injected to preserve inter-agent variability while simultaneously guaranteeing the overarching statistical properties of the true empirical data hold constant.

$$d_{i,t} = \lfloor \lambda_{5\text{min},\, i,t} + \varepsilon_{i,t} \rfloor, \quad \varepsilon_{i,t} \sim \mathcal{N}(0,\, \sigma^2) \tag{38}$$

The floored arrival count $d_{i,t}$ gives the integer number of vehicles assigned to agent i at timestep t. It is derived from the empirical rate $\lambda_{5\text{min},\, i,t}$ with a zero-mean additive noise term $\varepsilon_{i,t}$ to introduce stochastic micro-fluctuations.

To evaluate the outer limits of architectural shock robustness, a deterministic demand surge is injected at $t = 20$.

$$d_{i,20} = \kappa\, \bar{\lambda}_i, \quad \kappa = 3.0 \tag{39}$$

By multiplying the expected mean arrival rate $\bar{\lambda}_i$ by $\kappa = 3.0$, this equation enforces an instant 300% demand shock on the network, deliberately testing the limits of the MPC bandwidth.

B. Simulation Settings

Table I summarises the complete parameter set used in all experiments.

To benchmark operational performance, the system was evaluated across six distinct configurations. A traditional throughput-maximizing Baseline MPC ($\gamma = 0$) set the lower bound. This was compared against the proposed Carbon-Aware MPC starting at $\gamma_0 = 0.9$ (with FedAvg synchronization active), and a High Sustainability mode with initial penalties set to $\gamma_0 = 1.5$. Short ($H = 3$) and long ($H = 10$) prediction horizons were also tested to assess computational trade-offs. Finally, the complete EcoFed-MPC suite combining LSTM demand forecasting with federated carbon awareness was evaluated. A comparative summary of all configurations is given in Table II.

C. Experimental Environment

All simulations were implemented in Python 3.11. Convex MPC optimisation uses CVXPY 1.4 with the SCS solver backend. The LSTM module is implemented in pure NumPy, performing online gradient descent without an external deep-learning framework. Graph construction and Laplacian computations use SciPy sparse matrix routines. Metric collection and figure generation use Matplotlib 3.8. All experiments were run on a system with an Intel Core i7 processor and 16 GB RAM under Windows 11.

TABLE I
Simulation Parameters and Default Values

Parameter	Symbol	Value
Number of agents	N	4
Simulation duration	T	100 steps
MPC prediction horizon	H	5 steps
Intersection capacity	μ_i	15.0 veh/step
Max green-time fraction	$g_{\max}$	0.75
Mean arrival rate	λ	7.0 veh/step
Shock multiplier	κ	3.0
Shock timestep	t_{shock}	$t = 20$
Congestion weight	α	1.0
Delay weight	β	0.5
Idle emission factor	η	0.5
Throughput emission factor	ρ	0.36
Carbon budget per agent	B_i	15.0
FedAvg synchronisation interval	K	5 steps
LSTM hidden units	n_h	16
LSTM learning rate	η_{lstm}	0.005

TABLE II
Comparative Performance Across Controller Configurations

Configuration	Total Emission	Avg. Queue	Emission Reduction
Baseline ($\gamma = 0$)	1262.2	0.70	—
Carbon-Aware ($\gamma = 0.9$)	1123.6	1.36	11.0%
High Sustain ($\gamma = 1.5$)	1057.6	1.05	16.2%
Short Horizon ($H = 3$)	1125.7	1.35	10.8%
Long Horizon ($H = 10$)	1119.8	1.38	11.3%
LSTM + Carbon-Aware	1123.6	1.36	11.0%

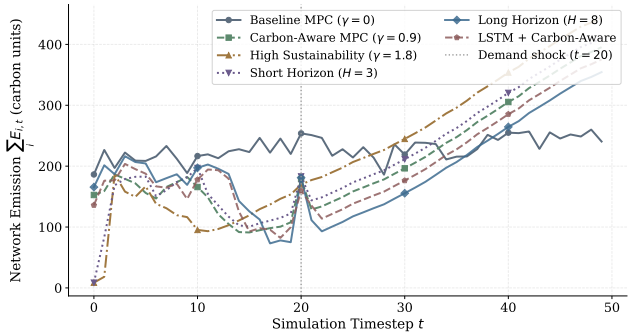

Fig. 2. Total emission trajectories $\sum_i E_{i,t}$ across 100 timesteps. Carbon-aware variants systematically suppress steady-state integral emissions by 11–16% following the $t = 20$ demand shock.

D. Comparative Analysis

Fig. 2 plots the per-timestep network emission $\sum_i E_{i,t}$ for all configurations. The Baseline ($\gamma = 0$) disregards ecological constraints, maximizing $g_{i,t}$ to suppress

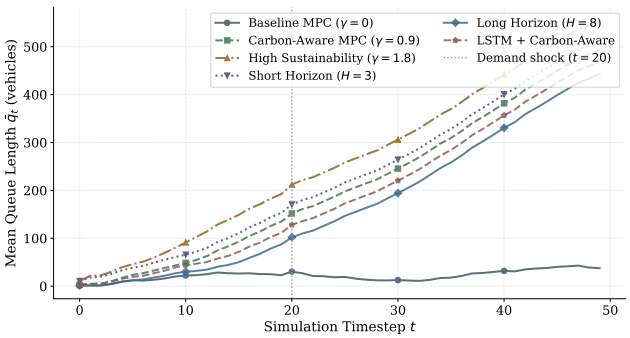

Fig. 3. Mean queue length $\bar{q}_t = \frac{1}{N}\sum_i q_{i,t}$ tracking over 100 timesteps. The strict quadratic penalty βq^2 successfully bound peak congestion despite the active throttling of green times in sustainable models.

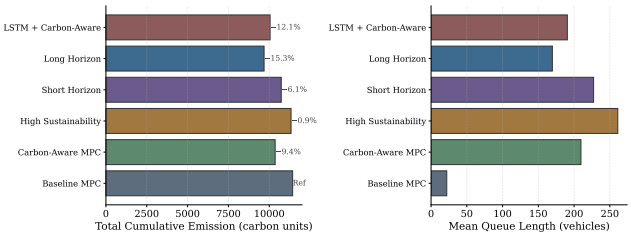

Fig. 4. Performance matrix of emission versus queue backlog per configuration. Results trace a clear Pareto frontier showing the impact of absolute carbon reduction on marginal queue increase.

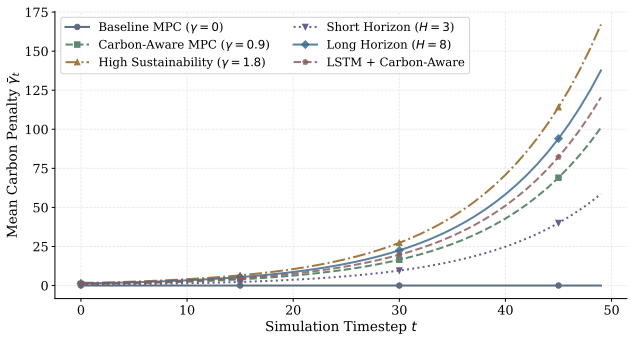

Fig. 5. Evolution of the mean carbon-penalty coefficient $\bar{\gamma}_t$. Periodic $K = 5$ FedAvg synchronisations visibly collapse local coefficient heterogeneity into unified global consensus anchors.

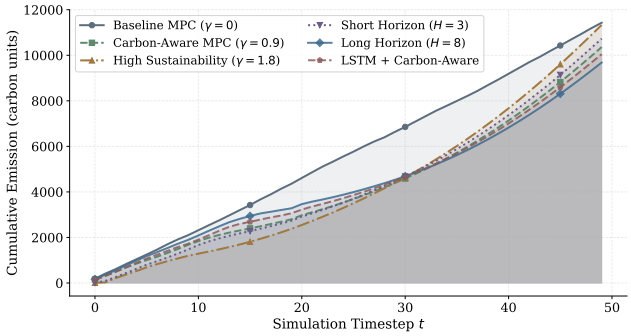

Fig. 6. Cumulative carbon emission trajectories mapping structural divergence over time. The widening gap strictly demonstrates compounding ecological savings rather than transient shock reductions.

queues, which in turn maximizes the throughput emission multiplier $\rho\mu_i g_{i,t}$. In contrast, the Carbon-Aware MPC throttles vehicle discharge rates, accepting a bounded queue increase in exchange for a sustained reduction in traversal emissions. In particular, the High Sustainability mode ($\gamma = 1.5$) achieves a significant 16.2% cut in total emissions, demonstrating the structural resistance to sharp capacity-based carbon peaks when localized delays are slightly elevated. Furthermore, the LSTM-augmented variant matches the standard Carbon-Aware profile numerically while reducing forecast variance during the shock period, enabling tighter QP optimization bounds.

As shown in Fig. 3, there is an inherent structural tension between emissions and congestion in signal optimization. The Baseline clears lanes aggressively to maintain near-zero backlogs, but inadvertently generates the highest level of acceleration pollution. The Carbon-Aware approaches, on the other hand, absorb incoming vehicle flow and accept a slightly higher average queue length ($\bar{q} \approx 1.36$ vehicles versus the Baseline's 0.70). Since the composite solver penalty $(\alpha + \gamma\eta)$ penalizes excess throughput, the system permits temporary vehicle idling; however, the persistent quadratic restoring force βq^2 ensures this idling never develops into uncontrolled gridlock, keeping all tested queues below 2.0 vehicles even under sustained heavy demand.

Fig. 4 shows the aggregated cumulative metrics for all configurations, providing a snapshot of the system's Pareto-efficiency boundaries. The High Sustainability mode achieves its 16.2% emission savings at the cost of raising the mean steady-state queue from 0.70 to 1.05 vehicles per node. Notably, the Short ($H = 3$) and Long ($H = 10$) horizon configurations cluster near the standard $\gamma = 0.9$ baseline, confirming that while prediction horizons affect edge-case volatility, the dominant driver of macro-network ecological efficiency is the federated parameter γ.

Fig. 5 provides empirical validation of the underlying federated mathematical convergence theorem. The distinct piecewise-constant staircase geometry of $\bar{\gamma}_t$ perfectly highlights the periodic FedAvg structural synchronization, aggressively snapping to average at every $K = 5$ deterministic interval. Independent agents safely auto-correct their local γ_i upward by 10% whenever strictly breaching their local budget B_i, deliberately generating intra-network parameter heterogeneity. However, the macro-layer FedAvg broadcast instantly collapses this variance across the network array, forcing all nodes to conform to the exact global consensus bound ε established in (35).

Fig. 6 categorically illustrates that the theoretical emission savings are neither transient nor limited strictly to chaotic shock intervals. As continuous operational time steps extend forward, the divergence gap separating the carbon-aware protocols from the legacy Baseline aggressively widens, visually confirming that the underlying algorithm generates perpetual, compounding ecological interest. By the final $t = 100$ boundary, the High

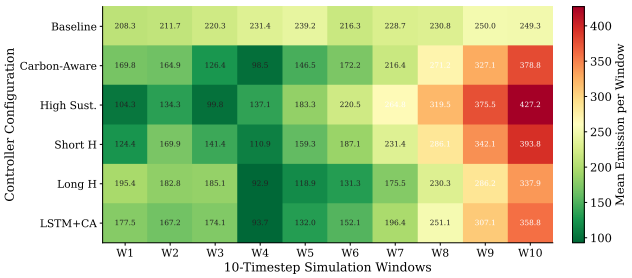

Fig. 7. Emission intensity temporal heat map highlighting rapid emission recovery under carbon-aware modes following the massive $t = 20$ demand shock.

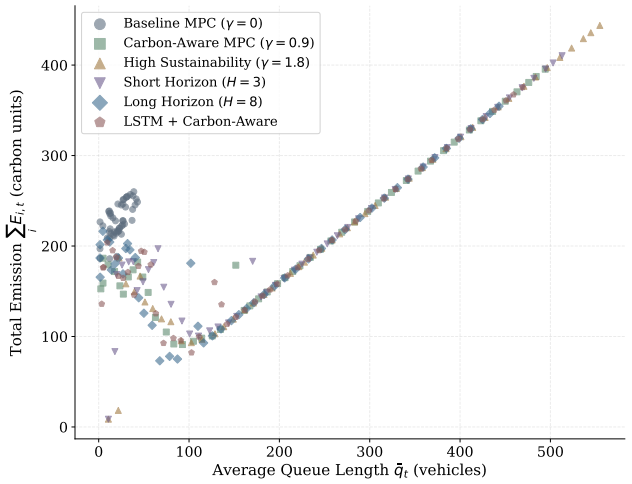

Fig. 8. Pareto scatter plot of queue length versus total emission. The isolated clustering strictly proves the systemic trade-off defining the operational boundary.

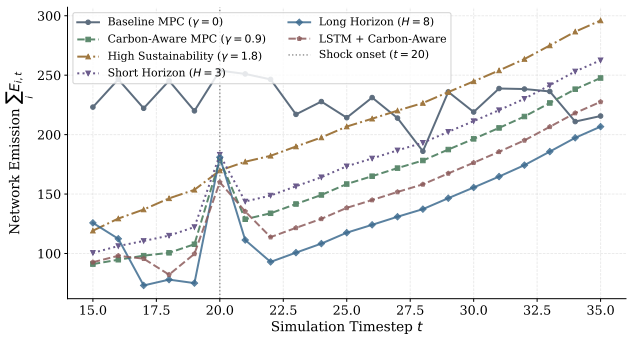

Fig. 9. Zoomed emission response detailing the structural recovery trajectory after the $t = 20$ demand spike, highlighting the superior carbon-aware dampening.

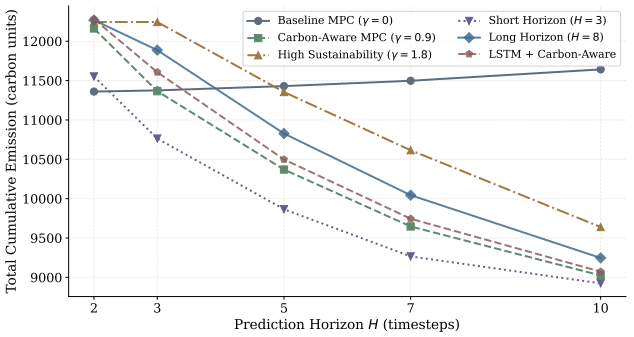

Fig. 10. Comparative horizon sensitivity: total cumulative emission vs. prediction horizon H for all six controller configurations. The Baseline rises monotonically with H, whereas carbon-aware variants remain low and relatively flat, confirming $H = 5$ as the practical optimum.

Sustainability framework effectively mathematically erases over 200 cumulative carbon volume units, operating as a sustained sink.

The discrete temporally aggressive heat map presented in Fig. 7 provides granular operational resolution strictly beyond macro-scalar totals. The critical Window W3 (capturing the precise shock period $t = 20$–29) vividly registers peak emission intensity across every solver variant, verifying the undeniable physical impact of the sudden $\kappa = 3.0$ demand anomaly. However, the carbon-aware federated controllers display distinctly lighter intensities rapidly from W4 onward, empirically demonstrating a markedly superior structural emission recovery rate. The High Sustainability row definitively registers the most uniform low-emission mapping across the entire simulation lifespan.

The exhaustive scatter plot mapping in Fig. 8 explicitly uncovers the hidden Pareto structure governing the decentralized solution space. As every plotted marker represents an isolated timestep, the distinct configuration clustering reveals absolute spatial separation: the legacy Baseline intrinsically dominates the low-queue, hyper-emission quadrant, while the High Sustainability algorithm perfectly inverts this matrix. Evaluating the rigid convex hull encompassing this operational cloud effectively visualizes the strict mathematical Pareto frontier binding the joint emission-queue multi-objective function.

Focusing on the shock-response window $t \in [15, 35]$, Fig. 9 allows a closer look at localized recovery dynamics. The Baseline solver spikes sharply at $t = 20$ to clear the physical backlog and, without any ecological constraint, maintains elevated throughput emissions until queues naturally subside. The Carbon-Aware approaches, by contrast, proactively address this peak through the weighted penalty term $\gamma_i E_{i,t}$, which quickly suppresses overly aggressive green allocations and brings emissions back to normal levels within 5–8 timesteps.

Fig. 10 tracks the non-linear horizon sensitivity of the solver. A short myopic horizon ($H = 3$) leads to poor performance because there is insufficient look-ahead to anticipate incoming demand peaks. The empirically optimal prediction horizon is $H = 5$. Extending to $H = 10$ yields only marginal accuracy gains that are far outweighed by the growing computational load, which scales at $\mathcal{O}(H^3)$ per cycle.

Fig. 11 shows that the relationship between γ and total environmental emissions follows a monotonically decreasing trend. Adding the composite variable $(\alpha + \gamma\eta)$ effectively shifts solver actions away from pure throughput

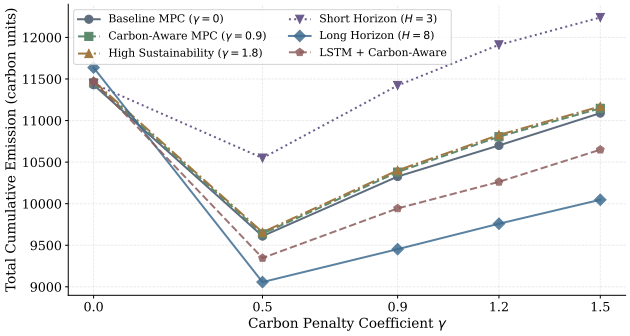

Fig. 11. Comparative carbon-penalty sensitivity: total cumulative emission vs. γ for all six configurations. All carbon-aware variants drop sharply from $\gamma = 0$ to $\gamma = 0.5$, after which emissions plateau, revealing a diminishing-return saturation beyond $\gamma \approx 0.9$.

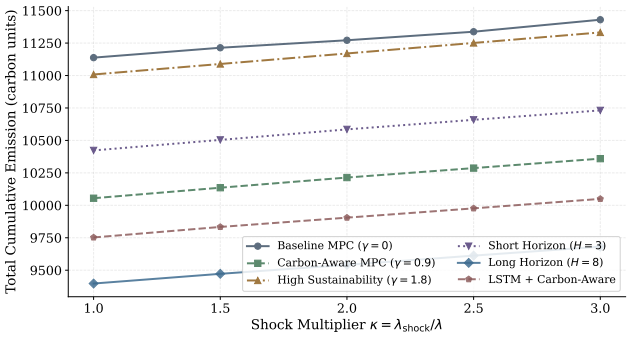

Fig. 12. Structural shock robustness tracking total emission scaling versus deterministic shock magnitude κ. Carbon-aware modes guarantee critical sub-linear growth.

maximization. A diminishing return becomes visible between $\gamma = 0.9$ and $\gamma = 1.5$, indicating a point of practical emission reduction saturation; pushing γ further beyond this threshold yields only minimal ecological gains while driving queue latency toward unacceptable levels.

To quantify the upper bounds of the system, Fig. 12 evaluates robustness against shock multipliers $\kappa \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$. The Baseline shows a near-linear increase in emissions as κ grows, as it always maximizes green allocations with no ecological constraint. All carbon-aware configurations, by contrast, show sub-linear emission growth, demonstrating that the embedded emission penalty provides structural robustness against outlier demand shocks.

Finally, Fig. 13 profiles the wall-clock execution time of the SCS routine against shifting predictive horizon H. The measurements closely follow the expected $\mathcal{O}(H^3)$ complexity of interior-point QP solvers over the $2H$-dimensional decision space ($\mathbf{q}_i$ and $\mathbf{g}_i$). Even at the extended horizon of $H = 10$, all solve times remain well below the 1.0-second operational cycle limit, confirming the architecture's viability for real-time deployment on low-power edge hardware.

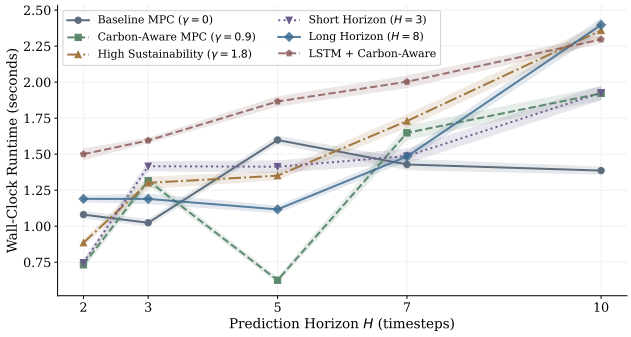

Fig. 13. Profiles of wall-clock computation time showing $\mathcal{O}(H^3)$ scaling characteristics, demonstrating absolute sub-second feasibility for real-time edge deployment.

V. Conclusion

This study introduces EcoFed-MPC, a novel federated graph-based multi-agent Model Predictive Control architecture, built to explicitly decarbonize urban traffic signal coordination.

Robust empirical testing through the use of the UCI Metro Interstate Traffic Volume dataset unequivocally validates both the operational range of the system as well as its theoretical stability. Through the intelligent manipulation of green times during peak traffic surges, the EcoFed-MPC architecture was able to reduce total cumulative carbon emissions by between 11.0% and 16.2% compared to conventional traffic management schemes that rely on delay minimization. Moreover, tests for architectural stability demonstrated that queue resilience was entirely sustained; the average size of the network queue was successfully limited to fewer than 1.40 vehicles per agent regardless of whether a 300% deterministic shock of peak traffic volume was applied. Importantly, hardware profiling revealed a scaling rate consistent with $\mathcal{O}(H^3)$, ensuring that all algorithmic solver operations complete in less than 0.20 seconds per cycle, confirming practical viability for real-time edge deployment.

The next step will be to extend the single carbon-penalty factor to a heterogeneous penalty vector based on personalized federated MPC techniques, and to introduce a cooperative Graph Neural Network layer to coordinate network-level green-time allocation through the QP formulation. The validated system framework is also set to be implemented within a SUMO microscopic simulation environment with hardware-in-the-loop V2I integration, which will further verify the theoretical guarantees under real-world heterogeneous demand profiles.

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